

Guy Blachar

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Research Interests

Probability and algebra, with particular emphasis on random walks on groups and graphs, probabilistic and geometric group theory, word maps, and stability questions in groups and algebras.

Academic Positions

Postdoctoral Fellow, Université Paris Dauphine-PSL February 2026–present
Mentor: Prof. Justin Salez.

Postdoctoral Fellow, Weizmann Institute of Science October 2024–January 2026
Mentor: Prof. Gady Kozma.

Research Visits

Visiting researcher, Mathematical Institute, University of Oxford 18–29 November 2024
Host: Prof. Emmanuel Breuillard.

Education

PhD in Pure Mathematics, Bar-Ilan University 2019–2024
Advisors: Prof. Gideon Amir and Prof. Uzi Vishne.
Dissertation: *The Law of Iterated Logarithm and Nonassociative Central Simple Algebras*.

MSc in Pure Mathematics, Bar-Ilan University 2015–2016
Advisor: Prof. Louis Rowen. Graduated summa cum laude.

BSc in Pure Mathematics, Bar-Ilan University 2012–2015
Graduated summa cum laude.

Awards and Fellowships

2025	Koshland Fellowship for Postdoctoral Researchers
2024	IMU Student Talk Prize
2024	Rothschild Prize Excellence Awards for PhD Students
2022–2023	The Efim Diner’s Award for Excellence in Math Teaching for Gifted Youth
2022–2023	Bar-Ilan University’s Outstanding Lecturer Award
2020–2021	Rector’s Prize for Outstanding Doctoral Students
2019–2023	Bar-Ilan University President’s Scholarship for Excellent Doctoral Students
2012–2013	Dean’s List

Publications

- [1] **The speed of random walks on semigroups** (with Be’eri Greenfeld), to appear in *Annales de l’Institut Henri Poincaré*.
- [2] **Probabilistic laws on infinite groups** (with Gideon Amir, Maria Gerasimova, and Gady Kozma), to appear in the *Israel Journal of Mathematics*.
- [3] **Mixability of finite groups** (with Gideon Amir, Subhajit Ghosh, and Uzi Vishne), *Random Structures & Algorithms* **67**, no. 4, e70037 (2025).
- [4] **Rank-stability of polynomial equations** (with Tomer Bauer and Be’eri Greenfeld), *International Mathematics Research Notices*, 2025, no. 13 (2025).
- [5] **Profinite rigidity of lamplighter groups**, *Proceedings of the American Mathematical Society* **153**, no. 9, 3735–3741 (2025).

- [6] **A law of iterated logarithm on lamplighter diagonal products** (with Gideon Amir), *Groups, Geometry, and Dynamics* **19**, no. 3, 1041–1080 (2025).
- [7] **Lipschitz harmonic functions on vertex-transitive graphs** (with Gideon Amir, Maria Gerasimova, and Gady Kozma), *Electronic Communications in Probability* **29**, 1–4 (2024).
- [8] **Semiassociative algebras over a field** (with Darrell Haile, Eliyahu Matzri, Edan Rein, and Uzi Vishne), *Journal of Algebra* **649**, 35–84 (2024).
- [9] **Identities of the algebra $\mathbb{O} \otimes \mathbb{O}$** (with Louis Rowen and Uzi Vishne), in *Noncommutative Rings and Their Applications VII: Algebra and Coding Theory*, Contemporary Mathematics **785**, 35–45 (2023).
- [10] **Abelianization of the Cartwright–Steger lattice** (with Orit Sela–Ben-David and Uzi Vishne), *Algebra and Discrete Mathematics* (2023).
- [11] **Noncommutative inclusion–exclusion, representations of left regular bands and the Tsetlin library** (with Louis Rowen and Uzi Vishne), *International Journal of Algebra and Computation* (2021).
- [12] **ℓ -weak identities and central polynomials for matrices** (with Eliyahu Matzri, Louis Rowen, and Uzi Vishne), in the proceedings of the workshop *Polynomial Identities in Algebras* (2021).

Preprints

- [1] **Galois groups of random polynomials of large degree** (with Emmanuel Breuillard and Gady Kozma); arXiv:2609.40123.
- [2] **Edge-regular graphs with non-negative curvature have polynomial growth** (with Hervé Pajot and Justin Salez), submitted; arXiv:2606.11094.
- [3] **Small entropy doubling for random walks and polynomial growth**, submitted; arXiv:2604.08490.
- [4] **Phase transition for recurrence of stationary random walks on lamplighter groups** (with Itai Benjamini and Ariel Yadin), submitted; arXiv:2510.06441.

Academic Service and Organization

- Member of the organizing committee, FPSAC 2021.
- Member of the organizing committee, *AI and Mathematics: Current Trends and Future Directions*, September 2024.
- Co-organizer, student seminar, Spring 2023.
- Co-organizer, mathematical programming competition, Summer 2023.
- Organizer of several reading groups in algebra and probability.
- Referee for several mathematical journals, including *Annals of Probability*, *Journal of Theoretical Probability*, *Journal of Algebra*, and *Mediterranean Journal of Mathematics*.

Talks

Conference Talks

- Algebra, Geometry, Dynamics, Bar-Ilan University, November 2025.
- IMU Annual Meeting, July 2025.
- Students Probability Day X, Weizmann Institute, May 2025.
- IMU Students Talks Day, September 2024.
- World of Groupcraft III, September 2023.
- IMU Students Talks Day, September 2023.
- 14th Ukraine Algebra Conference, July 2023.
- Students Probability Day IX, Weizmann Institute, May 2023.
- Poster session, *Groups of Dynamical Origins, Automata and Spectra*, March 2023.
- IMU Students Talks Day, September 2022.
- 2022 Brauer Group Meeting in Israel, June 2022.
- SageDays 110, October 2020.

Seminar Talks

- Internal Probability Seminar, Université Paris Dauphine–PSL, May 2026.
- HUJI Groups and Dynamics Seminar, January 2026.
- Haifa Cable Car Algebra Seminar, January 2026.
- Bar-Ilan Algebra Seminar, April 2025.
- Weizmann Probability Seminar, April 2025.
- European Non-Associative Algebra Seminar, February 2025.
- Algebra–Geometry–Combinatorics Seminar, TU Dresden, January 2025.
- Midrasha on Groups, Weizmann Institute, January 2025.
- TAU Groups and Dynamics Seminar, December 2024.
- Oxford Algebra Seminar, November 2024.
- Combinatorics, Probability, and Representation Theory Seminar, Indian Institute of Technology Madras, September 2024.
- TAU Probability Seminar, August 2024.
- HUJI Dynamics Seminar, May 2024.
- Probabilistic and Geometric Group Theory Online Seminar, May 2024.
- Bar-Ilan Probability Seminar, March 2024.
- HUJI Probability Seminar, February 2024.
- TAU Groups and Dynamics Seminar, February 2024.
- Bar-Ilan Algebra Seminar, January 2024.
- Bar-Ilan Algebra Seminar, January 2023.
- Ben-Gurion Probability and Ergodic Theory Seminar, December 2022.
- Technion Probability Seminar, November 2022.
- Bar-Ilan Probability Seminar, November 2022.

Teaching Experience

Instructor

- Introduction to Probability and Statistics for Financial Mathematics, Fall 2023.
- Linear Algebra I, Summers 2021–2024.

Teaching Assistant

- Introduction to Probability and Statistics, Springs 2021–2022.
- Group Theory, Falls 2022–2024.
- Rings and Modules, Springs 2021 and 2023.
- Fields and Galois Theory, Falls 2022–2024.
- Mathematical Probability, Springs 2021–2024.
- Linear Algebra I, Fall 2016.
- Algebraic Structures for Computer Science, Fall 2016 and Spring 2024.

Software and Technical Skills

- **SageMath interface to the GAP package GBNP** (with Tomer Bauer), available at <https://gitlab.com/mathzeta2/gbnp>.
- Proficient in Python and SageMath.